

NAME-DARSHAN PAWAR

USN-2VX23UE012

EXPT NO.6

TITLE-Numerical processing using numpy

```
In [8]: import numpy as np
arr=np.array([2,4,6,8])
print('org arr',arr)
arr_add=arr+2
arr_sub=arr-2
arr_mult=arr*2
arr_div=arr/2
arr_square=arr**2

print('arithmetic operations:')
print('add 2:',arr_add)
print('sub 2:',arr_sub)
print('mul 2:',arr_mult)
print('div 2:',arr_div)
print('squARE 2:',arr_square)
print()
arr_sqrt=np.sqrt(arr)
arr_exp=np.exp(arr)

print('universal operations:')
print("arr_sqrt ",arr_sqrt)
print("arr_exp",arr_exp)
print()

mean=np.mean(arr)
var=np.var(arr)
std_dev=np.std(arr)
arr_sum=np.sum(arr)

print('\nstatistics operations')
print('mean',mean)
print('std_dev',std_dev)
print('arr sum',arr_sum)
print('varience',var)

org arr [2 4 6 8]
arithmetic operations:
add 2: [ 4  6  8 10]
sub 2: [0 2 4 6]
mul 2: [ 4  8 12 16]
div 2: [1.  2.  3.  4.]
squARE 2: [ 4 16 36 64]

universal operations:
arr_sqrt [1.41421356 2.         2.44948974 2.82842712]
arr_exp [ 7.3890561  54.59815003 403.42879349 2980.95798704]

statistics operations
mean 5.0
std_dev 2.23606797749979
arr sum 20
arr sum 5.0
```

In []: